

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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TECHNICAL PROFILE

Low-level embedded firmware engineer developing real-time power, control, and communications software for microcontrollers and DSPs. Builds C/C++ firmware from peripheral bring-up and hardware debugging through updates, system integration, and validation, with supporting Embedded Linux experience.

TECHNICAL SKILLS

Low-Level Firmware: Embedded C/C++, bare-metal, FreeRTOS, TI HAL; peripheral bring-up, boot/update firmware

MCUs & DSPs: dsPIC33CK (16-bit C-Microchip HAL), PIC32 (32Bit FreeRTOS), TI AM62 (Embedded Linux), TMS320 DSP (C-TI HAL), STM32 (C-STM32 HAL), MSP430FR2355 (bare-metal C)

Communications: CAN 2.0/CAN FD; J1939, DroneCAN; I2C, SPI, UART/RS-232; IPMI/IPMB

Systems, Tools & Test: TI AM62 Embedded Linux, Yocto/Arago, kernel/device trees, systemd, Docker; Python, Bash, Git/GitLab, JTAG, KiCad, SocketCAN/PCAN, scopes/analyzers, professional through-hole/SMT soldering

ENGINEERING EXPERIENCE

AEGIS Power Systems

Embedded Firmware Engineer

May 2026–Present

- Develop and bring up 16-bit dsPIC33CK firmware, toolchains, and board peripherals one subsystem at a time; isolate register, timing, hardware, and configuration faults with targeted tests, scopes/analyzers, schematics, and datasheets.
- Implemented and validated Tier 2 management firmware for a VPX power supply, following VITA 46.11/IPMI 2.0 over I2C-based IPMB for chassis sensor, inventory, and event reporting.
- Integrate battery hardware, CAN-connected devices, and Linux applications in embedded power systems; trace failures from power and wiring through bus traffic, services, configuration, and code.
- Support TI AM62 Embedded Linux integration with Yocto/Arago, including kernel/device-tree changes, image deployment, boot diagnosis, systemd external-watchdog services, Docker, and Windows/Linux tooling portability.

Embedded Systems Intern

May–Aug 2025

Computer Science Intern

May–Aug 2024

Selected work across both internships

- Developed firmware for a two-processor power system: a PIC32 running FreeRTOS managed I2C, SPI, CAN/J1939, and UART communications, while a bare-metal TMS320 DSP handled real-time power control.
- Controlled custom transformer-coupled bidirectional buck-boost supplies from the DSP using 12 PWM A/B pairs (24 outputs); each A signal had configurable frequency, duty cycle, and relative phase, while B matched or inverted A.
- Built firmware-delivery tools for the same product: a C++ CLI accepted HEX files, while Python/TCP transport packetized updates with chunking and checksum validation; documented CLI, flashing, test, and toolchain workflows for engineering handoff.
- Implemented the device-side recovery path: the PIC32 staged firmware and a golden image in external SPI flash, programmed and verified the DSP over UART, and controlled boot.
- Created a restricted SSH/chroot command interface for TI Yocto/Arago power hardware and Bash/Python commands for secure remote inspection and control.
- Reverse-engineered RS-232 transactions and checksums for programmable electronic loads with limited documentation; helped build relay/load monitoring hardware and documented CLI, flashing, test, and toolchain workflows for engineering handoff.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Developed a repeatable Excel/VBA dimensional and spline-tolerance validation test that automatically flagged out-of-spec parts; built a searchable 17,000+ record tooling/part/vendor catalog with operation, location, and revision traceability.

SENIOR DESIGN

EV Active Sensor Adapter | ST MCU, TI BQ ADC, CAN

NC State University

- Helped architect an EV sensor adapter joining an STMicroelectronics MCU, TI BQ-family battery monitor/ADC, CAN transceivers, sensing, power, and telemetry on a custom PCB.
- Turned requirements and component trade studies into interface specifications and subsystem tests; coordinated design reviews, risk tracking, documentation, and bring-up across the team.

TEACHING & TECHNICAL LEADERSHIP

ECE 306 Teaching Assistant & Course Mentor | NC State University

Spring 2025–Spring 2026

- Supported three semesters: two in an unofficial role at about 10 hr/week and one as the official TA at 20 hr/week; delivered test preparation, labs, code/PCB debugging, project guidance, and ECE-program support.
- Taught students to build and debug MSP430FR2355 vehicle subsystems spanning ADC sensing, motor control, GPIO/interrupts, Wi-Fi/serial links, LCDs, and custom PCBs; added supplemental lectures and question-driven support as the class average rose from C+ the prior semester to A-.

IEEE Open Project Space 2 | Lead Instructor

Fall 2025 & Spring 2026

- Ran two semesters of a project-based electronics course from idea through demonstration; taught firmware, PCB/BOM workflows, and through-hole/SMT assembly while coaching students to solve their own design problems.

IEEE Soldering Workshops | Coordinator, Lecturer & Lab Instructor

- Led standalone workshops using custom through-hole and surface-mount flashing-light PCB keepsakes; taught professional technique, safety, inspection, debugging, and rework.